

# PLER: A Novel Full-Reference Metric for Evaluating the Geometric Fidelity of 3D Assets

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**Abstract**—This paper presents an enhancement of the previously proposed PLER (Peak Length to Error Ratio) metric for objective quality assessment of 3D models. The original PLER metric, which compares a reference and a test 3D model by analyzing the distances of rays cast from a bounding sphere, suffered from non-uniform ray distribution. To overcome this limitation, a comprehensive study of methods for generating uniformly distributed points on a sphere was conducted. The Fibonacci (golden) spiral method was identified as optimal, leading to the development of PLER-2.0. The paper details the comparative analysis of point distribution methods, presents experimental results demonstrating the superiority of the Fibonacci spiral, and validates the improved metric on test 3D models, showing increased accuracy and stability.

**Keywords**—3D models, quality assessment, full-reference metric, geometric fidelity, point distribution on sphere, Fibonacci spiral.

## I. INTRODUCTION

In our previous research [1], we introduced PLER (Peak Length to Error Ratio), a novel full-reference metric for evaluating the geometric fidelity of 3D models. This work continues the systematic investigation of quality assessment methods being developed at the MTUCI [2], building upon established approaches to image and video quality evaluation. The core principle of PLER involves placing both the reference and the test model within a sphere and casting rays from uniformly distributed points on the sphere's surface towards its center. The distances from the sphere surface to the first intersection with each model are recorded and compared, resulting in a quantitative quality score.

While the initial PLER metric proved functional for tasks not requiring high computational precision, it exhibited several shortcomings, primarily non-uniform distribution of rays across the model surface. This work is devoted to overcoming this key limitation. The process of enhancing the metric necessitated the selection of a method capable of generating a perfectly uniform distribution of points (ray sources) on the sphere's surface. This article demonstrates the results of the conducted research and its analysis, which led to the selection of the optimal method – the Fibonacci (golden) spiral. This

advancement brings the PLER metric to a qualitatively new level, hence the enhanced version is designated as PLER<sub>2.0</sub>. The previous version remains operational for less demanding applications.

## II. PROPOSED METHOD FOR 3D MODEL QUALITY ASSESSMENT

The PLER metric is based on comparing a reference and a test 3D model, both enclosed within a sphere. Rays are emitted from uniformly distributed points on the inner surface of this sphere towards its center. The distances from the sphere surface to the first intersection with the model are recorded, characterizing the difference between the reference and the test model.

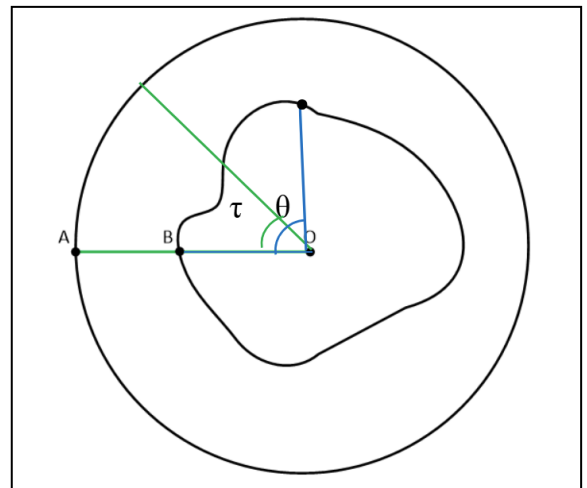


Fig. 1. Demonstration of the proposed method's principle.

The formula for PLER is given by:

$$PLER = 10 \lg \frac{[1 - L_{\min}(\varphi, \theta)]^2}{MSE} \quad (1)$$

where:

$1-L_{min}(\varphi, \theta)$  is the maximum distance from the center of the sphere to the surface of the reference model;

MSE is the root—mean-square error of the differences in the corresponding lengths for the reference and studied models.

For each point of the ray source, you need to set the direction (a point on the sphere) and the length of the ray (the radius value).

The construction of the direction (points on a unit sphere) is implemented by various methods of uniform distribution of points on the surface of the sphere, listed below. Each of the methods defines the distribution of unit vectors  $n$  with density:

$$fN(n) = \frac{1}{4\pi}, \|n\| = 1 \quad (2)$$

guarantees the same probability of hitting any element of the area  $\sin\varphi \, d\varphi \, d\theta$ .

For discrete sets of rays, the Fibonacci spiral method (point III,G) is convenient for uniform distribution of points on a sphere.

Advantages of the metric:

- **Scale Adaptability:** Automatic normalization via the sphere's radius makes the metric invariant to the absolute size of the models.
- **Selective Sensitivity:** The logarithmic scale enhances differences in the region of low-quality values, corresponding to psycho-physical perception characteristics.
- **Statistical Robustness:** Using MSE ensures robustness to outliers compared to metrics based on maximum error.

This approach aligns with established practices in camera system characterization, where statistical methods have proven effective for handling measurement variations [4].

Disadvantages of the metric:

- **Topological Blindness:** The metric does not detect topology changes (e.g., appearance/disappearance of holes) if geometric deviations are minimal.
- **Texture Insensitivity:** It considers only geometry, completely ignoring textures and materials.
- **Sampling Artifacts:** With a small number of rays, false matches are possible for models with a regular structure.

#### A. Mitigating Topological Blindness

To address topological blindness, an additional parameter is introduced – the Topological Similarity Index (TSI):

$$TSI = \frac{N_{ref} \cap N_{test}}{N_{ref} \cup N_{test}} \quad (3)$$

where  $N_{ref}$  and  $N_{test}$  are the counts of connected components and cycles in the reference and test models, respectively. A modified formula can be used:

$$PLER_{2.0} = w_1 * PLER + w_2 * TSI \quad (4)$$

where  $w_1 + w_2 = 1$ .

The TSI component has been implemented and tested, as shown in the results section.

#### B. Accounting for Textures and Materials

A multispectral version of the metric is planned for future research. This involves adding rays of different wavelengths (RGB) and performing a separate assessment for each color channel:

$$PLER_{color} = \frac{PLER_R + PLER_G + PLER_B}{3} \quad (5)$$

The proposed extension draws upon previously developed methods for color representation and analysis at MTUCI, including techniques for pseudo-color visualization in equal-contrast color space [5,6] and studies of color rendering characteristics [2].

### III. METHODS FOR GENERATING UNIFORMLY DISTRIBUTED RAY DIRECTIONS

A key element of the model is the generation of a set of unit vectors  $n_i$  that uniformly cover the sphere. The problem of generating uniformly distributed directions is closely related to broader research on random sequence generation for measurement and testing applications [3]. In the PLER metric, the Fibonacci spiral method was selected based on a study comparing the following methods:

#### A. Rejection-Based Methods

**Rejection Sampling in a Cube:** Points are randomly generated in a cube  $[-R, R]^3$  and accepted if they fall within the inscribed sphere. This method, while simple, has a rejection rate of approximately 47.6% and is conceptually related to methods later refined by Marsaglia [11] for efficient sphere point picking.

**B. Inverse Transform in Spherical Coordinates** Spherical coordinates  $(\theta, \varphi)$  are generated from uniform random variables  $[u_1, u_2]$ :  $\theta = \arccos(1 - 2u_1)$ ,  $\varphi = 2\pi u_2$ . This method ensures uniform distribution through the correct handling of the solid angle element  $d\Omega = \sin\theta \, d\theta \, d\varphi$ , as derived in standard references on sphere point picking [10].

#### C. Methods Based on Normal Distribution

**Muller's Method:** A point is generated from a 3D normal distribution and normalized to the unit sphere. This approach, originally proposed by Muller [10], is universal for any dimension and remains widely used in computational statistics and machine learning applications requiring uniform spherical samples.

#### D. Disk Projection Methods

Marsaglia's Method: Points are generated uniformly in a unit disk and then projected to the sphere. This method is based on the rejection approach proposed by Marsaglia [11], provides a probability of acceptance of  $\pi/4 \approx 0.7854$ , and is particularly effective for generating points on a sphere in three dimensions.

Two Uniforms Method: The z-coordinate and azimuth  $\phi$  are generated uniformly, followed by trigonometric calculations.

#### E. Generators Based on Random Rotations

Random Orthogonal Rotation: A random orthogonal matrix (via QR decomposition) is applied to a fixed vector. This method is computationally expensive.

Random Quaternion: A unit quaternion is generated uniformly and applied to a vector. This method is efficient in graphics engines.

#### F. Quasi-Random Sequences

Sobol Sequence: A low-discrepancy sequence in a cube, transformed to the sphere. For applications requiring hierarchical resolution and equal-area properties, specialized pixelization schemes such as HEALPix [12] offer alternative deterministic approaches, though they are optimized for specific resolutions rather than arbitrary N.

Latin Hypercube Sampling (LHS): Each coordinate is divided into intervals, ensuring even projection.

#### G. Deterministic Sets

Saff-Kuilaars Spiral: A spiral points method aiming for  $O(N^{-\frac{1}{2}})$  point separation. Saff and Kuilaars [13], who provided fundamental information about the minimum energy configurations and the properties of uniform distribution, which are the basis for modern spiral-based methods, studied the theoretical foundations of point distribution on a sphere.

Lebedev Quadrature: Predefined sets of points with exact weights for integration. Optimal for specific N but not arbitrary.

Fibonacci (Golden) Spiral: For index  $i = 0, \dots, N-1$ :

$$z_i = 1 - \frac{2i+1}{N} \quad (6)$$

$$\phi_i = \frac{2\pi i}{\phi}, \text{ where} \quad (7)$$

$$\phi = \frac{1+\sqrt{5}}{2} \text{ is golden ratio} \quad (8)$$

$$n_i = (\sqrt{1-z_i^2} * \cos \phi_i, \sqrt{1-z_i^2} * \sin \phi_i, z_i) \quad (9)$$

This method guarantees equal-area strips, uses an irrational step for even distribution, and has  $O(N)$  complexity.

The choice of deterministic uniform distribution is further justified by research on human visual system perception, which demonstrates the importance of controlled sampling patterns for reliable quality assessment [8].

H. In the experiment, the Fibonacci spiral showed the best values according to most statistical criteria.

The choice of (6) guarantees the same ring area (according to the formula  $\delta \cos \theta = 2/N$ ).

The azimuthal step  $\alpha = 2\pi/\phi$  uses the golden ratio  $\phi$ , its infinite chain splitting [1;1,1,1,...] leads to the best diophantine approximation, that is, to the minimum number of "close" points.

Thus, the Fibonacci spiral combines:

Determinism (a fully reproducible set for a fixed N).

The primary complexity of  $O(N)$  is only one pass through the array and three calls to  $\sin/\cos$ .

The best variance among all the considered methods (see the table above).

Therefore, in this paper, the Fibonacci spiral is chosen as the basic method of ray generation for calculating the PLER metric.

#### IV. COMPUTATIONAL EXPERIMENT ON UNIFORM DISTRIBUTION

A. Experiment Setup Sample sizes of  $N = \{100, 250, 500, 1000, 2500, 5000, 10000\}$  points were generated for each scheme, following the methodological recommendations for statistical validation in imaging systems [9] and building upon classical studies of spherical point distributions [10,11]. Metrics for uniformity were calculated.

##### B. Metrics for Assessing Ray Uniformity

$\chi_{\phi^2}, \chi_{\cos^2}$ : Chi-squared deviations for azimuth  $\phi$  and  $\cos(\theta)$ .

$KSD_{\phi}, KSD_{\cos}$ : Kolmogorov-Smirnov distances.

$CD_{\alpha}$ : Average absolute difference in point counts within spherical caps of angle  $\alpha$ .

$d_{geo}$ : Average geodesic distance to the nearest neighbor.

STD  $d_{geo}$ : Standard deviation of geodesic distances.

Hull Area (HA): Area of the convex hull of the point set (deviation from  $4\pi$ ).

C. Summary Results The Fibonacci spiral consistently demonstrated the best values across all metrics ( $\chi^2$ , KSD,  $CD_{\alpha}$ ,  $d_{geo}$ ,  $|HA - 4\pi|$ ) for all sample sizes, from  $N=100$  to  $N=10000$ .

D. Measurement Results Table I provides a summary of key metrics for  $N=10000$ , illustrating the performance of different methods.

TABLE I. SUMMARY OF UNIFORMITY METRICS FOR N=10000

| Method            | Mean $\chi$ | Mean KSD       | CD <sub>10</sub> | STD $d_{\text{geo}}$   | HA-4 $\pi$    |
|-------------------|-------------|----------------|------------------|------------------------|---------------|
| Rejection         | 11.17       | 0.02627        | 0.00232          | 0.0289                 | 0.0756        |
| Spherical Coords  | 15.64       | 0.01962        | 0.00213          | 0.0291                 | 0.0752        |
| Muller            | 15.65       | 0.03573        | 0.0021           | 0.0290                 | 0.0740        |
| Marsaglia         | 14.98       | 0.03162        | 0.00232          | 0.0296                 | 0.0740        |
| Two Uniform       | 11.29       | 0.02784        | 0.00213          | 0.0294                 | 0.0752        |
| Lambert           | 185.39      | 0.09039        | 0.00376          | 0.0311                 | 0.1405        |
| Random Rotation   | 373.42      | 0.22931        | 0.00597          | 0.0261                 | 1.6217        |
| Random Quaternion | 9.335       | 0.02171        | 0.00208          | 0.0288                 | 0.0786        |
| Sobol             | 0.23        | 0.00220        | 0.00089          | 0.0206                 | 0.0525        |
| Latin Hypercube   | 0.08        | 0.00099        | 0.00189          | 0.0281                 | 0.0709        |
| <b>Fibonacci</b>  | <b>0.08</b> | <b>0.00091</b> | <b>0.00068</b>   | <b>0.0011</b>          | <b>0.0390</b> |
| Lebedev           | 13.86       | 0.18898        | 0.01378          | 1.11*10 <sup>-16</sup> | 2.6518        |
| Spiral            | 1.04        | 0.00397        | 0.00059          | 0.0027                 | 0.0388        |

E. Graphical Representations and Interpretation Analysis of all metrics, including CD <sub>$\alpha$</sub>  (Fig. 2), box-plots of geodesic distances (Fig. 3), and empirical distribution functions, confirmed that the Fibonacci spiral provides the most uniform coverage across all tested values of N.

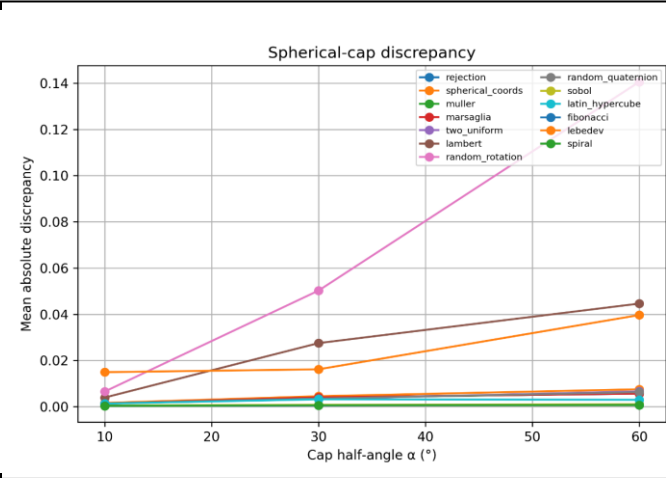
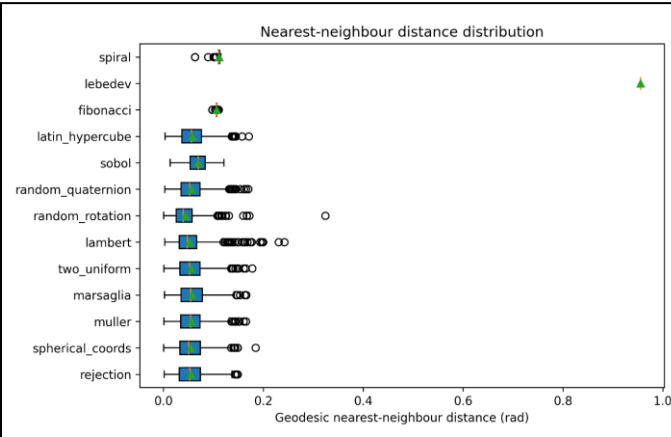
Fig. 2. Comparison of average CD <sub>$\alpha$</sub>  for  $\alpha = 10^\circ, 30^\circ, 60^\circ$ .

Fig. 3. Box-plot distribution of geodesic distances to the nearest neighbor.

## V. EVALUATION OF THE PLER METRIC ON TEST 3D MODELS

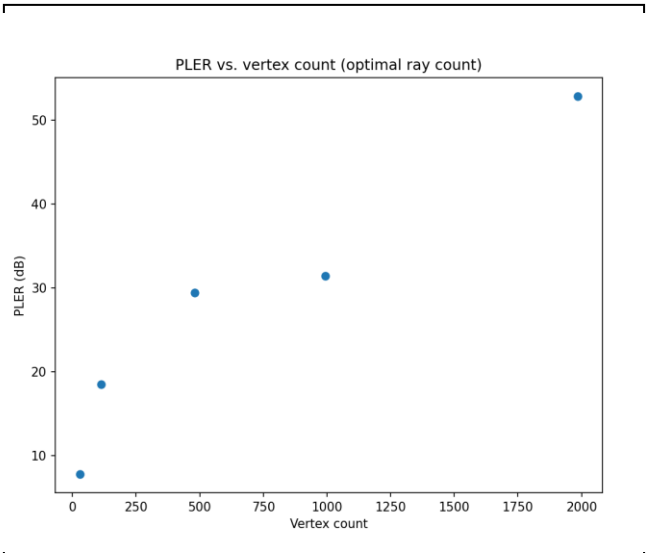
To validate the enhanced PLER<sub>2.0</sub> metric, a set of test UV-sphere models was used. UV-spheres were chosen as they provide a standardized geometric base with known properties, allowing for a clear assessment of the metric's behavior without the confounding influence of complex topology.

Ray projection parameters remained unchanged from the base PLER method, except for the range of ray counts, which corresponded to the uniformity experiment (N from 100 to 10000). The ray generator was fixed to the Fibonacci method.

For each model, the following were computed: PLER (dB), MSE, and TSI. The results are presented in Table II. Understanding such geometric fidelity metrics is increasingly important for applications involving AI-based object recognition, where model quality directly impacts detection performance [7]. As the number of rays N increases, the PLER value stabilizes and increases, indicating higher assessment accuracy. The TSI component provides complementary information about topological fidelity.

TABLE II. PLER (DB) AND TSI FOR TEST UV-SPHERE MODELS AT DIFFERENT RAY COUNTS

| Rays      | 100    | 500    | 1000   | 5000   | 10000  |
|-----------|--------|--------|--------|--------|--------|
| Sphere 1  | 7.5238 | 7.5196 | 7.5192 | 7.5202 | 7.520  |
| Sphere 2  | 17.319 | 17.343 | 17.346 | 17.345 | 17.345 |
| Sphere 3  | 25.811 | 25.787 | 25.790 | 25.791 | 25.791 |
| Sphere 4  | 27.006 | 27.001 | 26.999 | 27.002 | 27.003 |
| Sphere 5  | 34.146 | 34.097 | 34.091 | 34.096 | 34.099 |
| Sphere 6  | 34.837 | 34.816 | 34.813 | 34.819 | 34.821 |
| Sphere 7  | 44.028 | 44.035 | 44.024 | 44.023 | 44.024 |
| Sphere 8  | 44.615 | 44.549 | 44.567 | 44.561 | 44.562 |
| Sphere 9  | 76.824 | 77.188 | 77.121 | 76.904 | 76.844 |
| Sphere 10 | 100    | 100    | 100    | 100    | 100    |

Fig. 4. Graph of PLER(dB) vs. number of rays N for each model. Fig. 5. Error distribution curves  $|L_{\text{ref}} - L_{\text{disl}}|$  for N=2000.

## VI. CONCLUSION

A comparative analysis of the generators showed that the Fibonacci spiral (golden angle) achieves the smallest  $\chi^2$ , KS-distances and cap-discretion among all implemented methods. Its simplicity ( $O(N)$  arithmetic + one sin/cos call) and determinism make it preferable in tasks where precise sphere coverage is required (AR/VR rendering, photometric verification, 3D model quality assessment).

The PLER<sub>2.0</sub> metric, based on rays distributed along a Fibonacci spiral, demonstrates:

Monotonous growth of PLER<sub>2.0</sub> with a decrease in the number of vertices of the model (Table 1).

Low miss rate ( $< 0.1\%$  even at 100 rays) due to the full coverage of the sphere.

The execution time ( $< 6$  seconds per 1000 beams) and memory consumption ( $< 200$  MB) allow PLER to be used online (for example, in a real-time quality assessment system).

The prospects:

- Adaptive selection of the number of rays can reduce calculations for simple models without loss of accuracy.
- The integration of textural and illuminated components (extending PLER to PLER<sub>color</sub>, PLER<sub>vis</sub>) will open up the possibility to evaluate not only geometry, but also visual perception.
- Training neural networks based on Ray-Depth distance arrays (Ray-Depth Maps) can provide even faster predictive quality assessment.

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